

Depth-Staggered Fibonacci Spacing for Sparse Attention: Static Schedules Beat Learned Dilation and Extrapolate Where Dense Attention Fails*

Chad A. Capps
Independent Researcher
Chad.Capps@me.com

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Abstract

We study sparse self-attention in which each query attends to a dense local window plus a set of Fibonacci-spaced offsets, with a per-layer scalar α that compresses or expands the spacing. Across 21 language models trained under one matched recipe (60M parameters, 512 hidden, 16 layers, 426M tokens), we compare four ways of setting α across depth: fixed, per-layer learned, a static linear stagger, and a coprime (anti-gridding) reassignment of that stagger, together with a reach-matched power-of-2 control. Three results stand out. First, a static per-layer stagger improves perplexity over both fixed and learned α , and the gain is base-agnostic: applying the same stagger to a power-of-2 base lifts it above fixed Fibonacci and to parity with learned Fibonacci attention. Second, learning α per layer is inert: it does not beat the static schedule and costs roughly five times the inference latency. Third, and most consequential, all sparse variants extrapolate to four times their training length with little or no degradation, whereas a recipe-matched dense baseline collapses (perplexity rises by 201% at $4\times$ length); we attribute this to fixed-offset attention only ever querying relative positions seen during training. We also report two honest negatives: at training length the best sparse model has about 26% higher perplexity than the dense baseline, and the staggering gain is uniform across context positions rather than concentrated at long range.

1 Introduction

Sparse attention reduces the quadratic cost of self-attention by restricting each query to a subset of keys. A recurring design question is *which* subset, and in particular how the chosen offsets should vary with depth. Many schemes fix one pattern and rely on composition through the stack to widen the receptive field; others vary the pattern across heads. Comparatively little is known about varying a single spacing parameter *per layer* in a controlled language-modeling setting.

We take the offsets to follow a Fibonacci sequence (denser at short range, geometrically sparser at long range) and attach a per-layer scalar α that scales every offset, $d_k = \alpha f_k$. The original motivation was a “spring”: let each layer *learn* its own α so that depth-appropriate spacing emerges during training. That hypothesis is not supported by our data. The learned scalar barely moves from its initialization and yields no advantage over a fixed schedule. What does help is setting α

*Code, the paper source, and the experiment database are available at <https://github.com/ccapps42/scaled-fibonacci-attention>.

statically but differently per layer, a fixed stagger across depth that we imposed rather than learned. This paper reports the full arc, including the negative result on learning, because the static schedule that wins is simpler than the mechanism we set out to study.

Our contributions are:

- A controlled comparison of four per-layer α schedules (fixed, learned, static linear stagger, coprime stagger) and a reach-matched power-of-2 control, all sharing one attention implementation and one training recipe, over 21 trained models and a window sweep $W \in \{6, 8, 10, 12\}$ (Section 4).
- Evidence that a static per-layer stagger is the dominant lever: it beats both fixed and learned α , and it is *base-agnostic*: staggering a power-of-2 base raises it above fixed Fibonacci and to parity with learned Fibonacci attention. The Fibonacci base then adds a separable gain on top (Section 4.2).
- A length-extrapolation result: structured sparse attention is robust out to $4\times$ training length while a recipe-matched dense baseline collapses, with a mechanism (no out-of-distribution relative positions) that also explains the ordering among sparse variants (Section 4.4).
- Two honest negatives that bound the claims: a training-length quality gap to dense, and a position-resolved analysis showing the staggering gain is uniform rather than long-range-specific (Section 4.5).

2 Method

Attended set. For a query at position i , the attended key set is the union of a dense local window and a set of scaled Fibonacci offsets. The window covers distances $\{0, 1, \dots, W\}$ (self-position included, so no query has an empty key set). The sparse offsets use the base sequence

$$\text{fib} = [1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987], \quad K = 15,$$

with each rung scaled to a target distance $d_k = \alpha f_k$ and gathered at the key/value rows $i - d_k$. Because d_k is continuous, we gather by linear interpolation between the floor and floor-plus-one positions, which keeps a nonzero gradient to α even when every rung coincides with an integer position. Rotary position embeddings are applied at true positions before gathering. Offsets are shared across all queries, so this is a set of shifted gathers rather than per-position index lists.

The per-layer scalar. We write $\alpha = 0.5 + \text{sigmoid}(\theta) \in [0.5, 1.5]$, with θ the only added parameter. The base sequence already spans the context at $\alpha = 1$ ($f_{15} = 987 \approx 1024$), so the useful dynamic range is compression ($\alpha < 1$) plus a small margin above 1. We study four ways of setting θ across the $L = 16$ layers:

- **Fixed:** $\alpha_\ell = 1$ for all layers.
- **Learned:** θ_ℓ is a free parameter per layer, initialized to $\alpha = 1$.
- **Staggered (linear):** $\alpha_\ell = 0.5 + \ell/(L - 1)$, a fixed ramp from 0.5 at layer 0 to 1.5 at layer $L - 1$.
- **Staggered (coprime / HDC):** the same multiset of α values, reassigned to layers by a coprime stride $\ell \mapsto (S\ell \bmod L)$ with $S = 7$, so adjacent layers receive maximally different spacing. This ports the anti-gridding idea from dilated convolutions (Wang et al., 2018) to per-layer attention spacing.

Control base. As a control we replace the Fibonacci base with a reach-matched power-of-2 base (log-sparse). It reaches the same maximum distance with fewer rungs, isolating the effect of spacing *density* at equal reach rather than rung count.

Implementation note. A hard boolean attention mask carries no gradient to α , so the learned variant uses the interpolated gather. Fixed and staggered variants have integer offsets and could use a fused boolean-mask path, but we report them through the same gather module to remove any numerical-path confound; for the length-extrapolation study the integer-offset variants additionally run through a fused boolean-mask attention at evaluation time, which is numerically identical and faster. All variants share projections, GQA shapes, and RoPE with the dense baseline; the only added parameters are the θ scalars.

3 Experimental Setup

Model and training. All models use the same architecture: $d_{\text{model}} = 512$, 16 layers, 8 attention heads, 2 key/value heads (GQA), head dimension 64, vocabulary 32,768, sequence length 1024, SwiGLU feed-forward with ratio 8/3, RMSNorm, RoPE ($\theta_{\text{rope}} = 10^4$), tied embeddings, no dropout. Training is identical across runs: 13,000 steps at an effective batch of 32,768 tokens (426M tokens total), AdamW ($\beta = 0.9, 0.95$, weight decay 0.1), learning rate 3×10^{-4} with 300 warmup steps and cosine decay, gradient clip 1.0, bf16 autocast, seed 42. The dense baseline (denoted DENSE) shares this recipe exactly; the only difference is its attention mechanism (verified by a field-by-field config comparison). Total non-embedding parameters are 44.1M (60.9M including the tied embedding).

Data and evaluation. Models train on a fixed token mixture, in approximate token proportions 35% FineWeb-Edu (Penedo et al., 2024), 22% Wikipedia (Wikimedia Foundation, 2023), 20% TinyStories (Eldan and Li, 2023), and 23% mathematics (the MATH dataset (Hendrycks et al., 2021) and OpenMathInstruct-2 (Toshniwal et al., 2024)). We evaluate by token-level perplexity on four held-out sets: FineWeb-Edu, Wikipedia, TinyStories, and a math set held out from OpenMathInstruct-2 (problem-solution pairs, a disjoint shard from the training math). We additionally report two synthetic multi-hop reasoning tasks (RULER variable tracking (Hsieh et al., 2024) and LEGO (Zhang et al., 2022)) and an analytic FLOPs/throughput breakdown. The sparse models add only the θ scalars over dense, so the comparison to DENSE is essentially parameter-matched.

Run matrix. We sweep the local window $W \in \{6, 8, 10, 12\}$ for each of: fixed Fibonacci, learned Fibonacci, linear-staggered Fibonacci, coprime-staggered (HDC) Fibonacci, and linear-staggered power-of-2. The plain power-of-2 control (fixed, $\alpha = 1$) is run at $W = 8$. This yields 21 trained models. Throughput note: the learned variant runs through the unfused interpolated gather and is roughly $5\times$ slower at inference than the fused integer-offset variants at identical FLOPs; we treat efficiency as context, not as a selection criterion, since at this shape the attention term is a minority of per-layer compute.

4 Results

4.1 Perplexity: staggering is the dominant lever

Table 1 reports held-out perplexity for the full matrix. Within the Fibonacci base, the ordering is consistent across every window and dataset:

$$\text{staggered} \approx \text{coprime} > \text{learned} > \text{fixed},$$

and the best model overall is the linear stagger at $W = 12$ (FineWeb 42.02, Wikipedia 37.59). The coprime reassignment ties the linear stagger to within seed-scale differences at every window, so the *cascade order* of the per-layer spacings does not matter; only the multiset of spacings does. The plain power-of-2 control is the weakest model in the matrix.

base / schedule	W	TinyStories	Math	FineWeb-Edu	Wikipedia
fib / fixed	6	5.57	3.45	44.35	40.24
fib / fixed	8	5.55	3.45	44.08	40.02
fib / fixed	10	5.48	3.42	43.48	39.33
fib / fixed	12	5.44	3.39	43.12	39.05
fib / learned	6	5.53	3.40	43.86	39.62
fib / learned	8	5.52	3.40	43.70	39.43
fib / learned	10	5.43	3.35	42.99	38.80
fib / learned	12	5.40	3.35	42.85	38.60
fib / stagger	6	5.39	3.28	42.93	38.59
fib / stagger	8	5.36	3.27	42.53	38.07
fib / stagger	10	5.33	3.26	42.21	37.88
fib / stagger	12	5.30	3.25	42.02	37.59
fib / coprime	6	5.40	3.29	43.10	38.70
fib / coprime	8	5.37	3.28	42.58	38.27
fib / coprime	10	5.33	3.27	42.28	38.00
fib / coprime	12	5.30	3.26	42.09	37.75
pow2 / fixed	8	5.67	3.55	44.78	40.89
pow2 / stagger	6	5.54	3.43	44.11	39.89
pow2 / stagger	8	5.48	3.41	43.60	39.56
pow2 / stagger	10	5.43	3.39	43.17	39.05
pow2 / stagger	12	5.38	3.36	42.69	38.51

Table 1: Held-out perplexity for all 21 models (lower is better). Within the Fibonacci base, staggered $\approx$ coprime $>$ learned $>$ fixed at every window. The best model is the linear stagger at $W=12$.

4.2 The staggering gain is base-agnostic, and the base adds to it

Staggering the power-of-2 base lifts it substantially: at $W = 8$, plain log-sparse improves from 44.78 to 43.60 FineWeb perplexity once the same linear α ramp is applied. The cleanest statement of the effect uses the three-seed means at $W=12$ (Table 4): *power-of-2 staggered (42.84) beats fixed Fibonacci (43.18) and matches learned Fibonacci (42.85)*. A worse base, merely staggered, equals or outperforms the better base whether it is fixed or learned. So per-layer spacing diversity dominates the base choice, and learning α buys nothing that staggering does not. (A single seed had suggested power-of-2 staggered edged ahead of learned Fibonacci by 0.16; across three seeds that gap vanishes to 0.01, within the seed-noise band of Table 4, so we report it as a match.)

The Fibonacci base nonetheless contributes a separable gain. At matched staggering, Fibonacci beats power-of-2 at every window (FineWeb: 42.93 vs. 44.11 at $W=6$ down to 42.02 vs. 42.69 at $W=12$). The two factors (staggering and base density) are additive rather than redundant.

Why staggering widens coverage. At fixed $\alpha = 1$ every layer attends the same offsets, so the set of distinct distances reachable anywhere in the stack equals one layer’s offsets (about 19 distinct distances beyond the window at sequence length 1024). A linear α ramp gives each layer a different scaled copy of the same rungs, raising the union of reachable distances by roughly $4\times$ (to about 75) at the same per-layer sparsity and the same attention FLOPs.

This count of distinct distances still understates the receptive field, because the rungs are not point samples. The dense local window of width W acts as a smoothing kernel at every layer: within a few layers each token’s representation already aggregates its $\pm W$ neighbors, and the blur compounds with depth. A rung read at distance $d_k = \alpha f_k$ therefore returns a soft band of half-width on the order of W centered on d_k , not a single position. Consecutive Fibonacci rungs are separated by $f_{k+1} - f_k = f_{k-1}$, so whenever that spacing falls below about $2W$ the bands of neighboring rungs overlap and the apparent gap between them is not a real hole. This is automatic at short range, where the rungs are dense (1, 2, 3, 5, 8, ...); the spacing only exceeds $2W$ far out (for example $610 \rightarrow 987$), where the precise offset matters least. Staggering reinforces the effect: because each layer scales the same nominal rung to a different absolute distance, the union over layers places several bands in an interval that any single layer would leave empty, and multi-hop composition adds the sums $d_a + d_b$ of rungs used in successive layers. The distinct-distance count is thus a conservative lower bound on coverage.

This argument explains why the scheme has no functional gaps, not where its measured advantage comes from. The position-resolved analysis in Section 4.5 shows the staggering gain is uniform across context positions rather than concentrated at long range, so the widened receptive field is better read as evidence that staggering sacrifices no coverage than as the source of the perplexity improvement.

4.3 Learning the scalar is inert

The learned schedule improves over fixed (Table 1) but never reaches the static stagger, and it does so at roughly $5\times$ the inference latency because the non-integer offsets require the unfused gather. The learned α values stay close to their initialization. The mechanism we set out to study, a trainable spring, is therefore dominated by a fixed schedule we imposed by hand. We read this as evidence that the benefit is structural (diverse per-layer spacing) rather than something gradient descent needs to discover.

4.4 Length extrapolation: sparse is robust, dense collapses

All models train at sequence length 1024. We rebuild each model at evaluation lengths 2048 and 4096 (RoPE and the offset set extend naturally; no length-dependent parameters exist) and measure perplexity. Table 2 shows the result. The structured sparse models are flat to slightly improving out to $4\times$ length. The dense baseline, best at training length, degrades catastrophically: FineWeb perplexity rises from 33.37 to 100.36, a 201% increase, and is already behind the sparse models by 2048.

Mechanism. Fixed-offset sparse attention only ever queries a bounded set of relative positions, nearly all of which appear during training. RoPE rotation angles for those offsets are therefore in-distribution at any sequence length. Dense attention, by contrast, attends every relative position,

model	$T=1024$	$T=2048$	$T=4096$	change at $4\times$
DENSE	33.37	49.15	100.36	+201%
fib / stagger $W=12$	42.02	41.65	41.45	-1.3%
fib / stagger $W=8$	42.53	42.17	41.98	-1.3%
fib / coprime $W=12$	42.09	42.53	43.25	+2.8%
fib / fixed $W=8$	44.08	43.74	43.56	-1.2%
fib / learned $W=8$	43.70	43.34	43.15	-1.3%
pow2 / fixed $W=8$	44.78	44.98	49.86	+11.3%

Table 2: FineWeb-Edu perplexity as a function of evaluation length for models trained at length 1024. Dense is best at 1024 but collapses past it; structured sparse attention is robust to $4\times$.

including distances between 1024 and 4096 whose rotary angles were never trained; vanilla RoPE without scaling does not extrapolate to them, and perplexity blows up. The same mechanism explains the ordering among sparse variants. Plain power-of-2 degrades more (+11.3%) because its reach-matched base introduces new large rungs (1024, 2048) at long evaluation length that were never trained; the Fibonacci base, with a fixed 15-rung set, introduces no such new offsets. The coprime variant degrades mildly (+2.8%), making the linear stagger the more length-robust of the two. The practical reading: at training length sparse attention pays a quality cost, but it extrapolates at near-zero cost to lengths where dense attention is unusable.

4.5 Honest negatives

Training-length gap to dense. At length 1024 the dense baseline reaches 33.37 FineWeb perplexity versus 42.02 for the best sparse model, about 26% better. Every sparse variant in the matrix is well above dense at training length. At this scale, sparse attention is a substantial quality sacrifice in-distribution; its value here is length generalization, not in-distribution quality.

The gain is not long-range-specific. We resolve loss by token position within the 1024 window (Table 3). If the staggering advantage came from wider long-range coverage, the gap to fixed attention should grow at later positions, where more distant context is available to reach. It does not: the staggered-minus-fixed gap on FineWeb is roughly flat at +0.035 nats across all but the first position bin. The perplexity improvement is a uniform, broadly distributed gain in modeling quality, not a long-range-reaching effect. The effective-receptive-field argument explains *why* sparse attention works at all; it does not explain *where* the staggering gain comes from, which appears to be general per-layer representational diversity.

schedule	b0	b1	b2	b3	b4	b5	b6	b7
fib / fixed $W=8$	3.883	3.808	3.767	3.770	3.742	3.780	3.776	3.763
fib / stagger $W=8$	3.861	3.769	3.733	3.731	3.709	3.737	3.739	3.726
gap (fixed - stagger)	.023	.039	.033	.040	.033	.044	.037	.037

Table 3: Position-resolved FineWeb NLL (nats), early context (b0) to late context (b7). The staggering advantage is flat across position bins, not growing at long range.

Reasoning evaluations are at chance. The two synthetic multi-hop tasks (RULER variable tracking, LEGO) score at or near chance for every model, including the dense baseline. At 60M

parameters these tasks are capacity-limited, so they do not discriminate among attention patterns here; we report them for completeness and rest the quality claims on perplexity and the extrapolation result.

4.6 Robustness to seed

The full matrix uses a single seed. To bound the seed noise and re-test the smallest margins, we retrained the five $W=12$ configurations at additional seeds—three seeds each for the four integer-offset (bool-mask) configs, two for the learned config, which is far costlier to train. Table 4 reports the per-config mean and standard deviation. The seed noise is small on FineWeb (per-config std 0.06–0.18, pooled ≈ 0.13), so the three headline margins clear it comfortably: staggered beats fixed by 1.10 ($\sim 8\sigma$), beats learned by 0.77 ($\sim 6\sigma$), and beats power-of-2 staggered by 0.76 ($\sim 5\sigma$). The two near-ties are confirmed as ties: staggered versus coprime is 0.10, smaller than either config’s std and with overlapping seed ranges; power-of-2 staggered versus learned Fibonacci is 0.01.

config ($W=12$)	seed 42	seed 43	seed 44	mean $\pm$ std
fib / stagger	42.02	41.96	42.26	42.08 ± 0.13
fib / coprime	42.09	42.03	42.43	42.18 ± 0.18
pow2 / stagger	42.69	42.77	43.08	42.84 ± 0.17
fib / learned	42.85	42.85	—	42.85 ± 0.00
fib / fixed	43.12	43.16	43.27	43.18 ± 0.06

Table 4: Seed replication at $W=12$ (FineWeb-Edu perplexity). Four integer-offset configs at three seeds, the learned config at two. Per-config std is 0.06–0.18; the three headline margins exceed it by 5–8 $\times$, while staggered-vs-coprime and pow2-staggered-vs-learned are within it (ties).

5 Related Work

Fixed and per-head sparse patterns. Strided and log-sparse patterns (Child et al., 2019), local-plus-global windows (Beltagy et al., 2020; Zaheer et al., 2020), and geometric dilation mixed within layers (Ding et al., 2023) widen the receptive field through composition or across heads. PowerAttention (Chen et al., 2025) frames coverage through the stack explicitly but uses the same pattern in every layer. Fibottention (Rahimian et al., 2024) uses Fibonacci/Wythoff dilation that differs across heads and is deterministic and non-adaptive. Our staggering differs by varying a single spacing scalar *per layer* and by isolating that choice in a matched language-modeling comparison.

Per-layer spacing schedules. WaveNet (van den Oord et al., 2016) stacks per-layer increasing dilation for an exponential receptive field at constant parameters; the gridding analysis of Wang et al. (2018) shows that common-factor dilation rates leave periodic gaps and that coprime rates avoid them. Our coprime variant ports that fix to attention; we find it does not improve language-modeling perplexity over a plain linear ramp at this scale, and is slightly less robust under length extrapolation. DilateFormer (Jiao et al., 2023) varies dilation per head rather than per layer. Closest to our setting, MSWA (Xu et al., 2025) staggers the *local window size* across both heads and layers, widening it from shallow to deep layers for broader receptive field through the stack. The distinction is the quantity being staggered: MSWA varies the contiguous window *width*, whereas we hold the local window fixed and stagger the *sparse Fibonacci dilation spacing*: the offsets at which each layer samples beyond the window.

Learned spacing. Adaptive attention span (Sukhbaatar et al., 2019) learns a per-head/per-layer scalar controlling a contiguous attention reach via a soft-mask cutoff; that parameter trains cleanly. Dilated convolution with learnable spacings (Khalfaoui-Hassani et al., 2023) learns spacing in convolutions. In contrast to the span-cutoff case, our learned multiplicative dilation scalar is inert (it does not move appreciably and does not beat a fixed schedule), which we attribute to a flat loss landscape in the dilation multiplier.

Length extrapolation. ALiBi (Press et al., 2021) biases attention by distance to extrapolate, and RoPE (Su et al., 2021) is the position scheme we use. Our extrapolation result is not a new positional method but an observation: fixed-offset sparsity confines attention to trained relative positions, so it inherits robust extrapolation from RoPE at no extra cost, while dense attention with the same RoPE does not.

6 Discussion and Limitations

The headline practical finding is a trade: structured sparse attention costs in-distribution quality at training length but extrapolates to several times that length where a recipe-matched dense model with the same positional scheme fails. The within-sparse design lessons are that a static per-layer stagger beats both a fixed schedule and a learned one, that the effect is base-agnostic, and that the Fibonacci base adds a separable gain.

Several limitations bound these claims. The full matrix uses a single seed; we replicated the five $W=12$ configurations at two to three seeds (Section 4.6) and find the three headline margins exceed the seed noise by 5–8 $\times$, while the two near-ties are within it. The remaining windows ($W=6, 8, 10$) are still single-seed, and the consistency of the learned-versus-staggered and staggering-versus-fixed gaps across windows and datasets is supporting rather than independent evidence at those windows. The scale is 60M parameters, where the reasoning tasks are at chance and the absolute perplexities are high; whether the extrapolation advantage and the staggering gain persist at larger scale is open. The extrapolation result is specific to vanilla RoPE without scaling; length extrapolation methods built for dense attention would narrow or close the dense collapse, and the fair comparison at that point is an open question. Finally, the staggering schedule (a linear ramp over $[0.5, 1.5]$) was fixed by design and only lightly varied (the coprime reassignment); the optimal per-layer schedule is not characterized here.

7 Conclusion

Across 21 matched language models, a static per-layer spacing stagger is the most effective way to set sparse-attention offsets we tested: it beats a fixed schedule, beats a learned per-layer scalar, and the gain transfers to a different offset base. The mechanism we originally set out to study, a learned spring, is inert. The result with the widest implication is that fixed-offset sparse attention extrapolates to 4 $\times$ training length at near-zero cost while dense attention under the same positional scheme collapses, because sparsity confines attention to relative positions seen during training. We report alongside these the honest negatives that bound them: a training-length quality gap to dense and a position-resolved analysis showing the staggering gain is broad rather than long-range-specific.

Use of Generative AI

The author used a generative AI language model (Anthropic’s Claude) to assist with drafting and editing the manuscript and with writing the training, evaluation, and analysis scripts. All AI-assisted output, including every claim, numerical result, and reference, was reviewed and verified by the author, who takes full responsibility for the content of this paper.

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